



Do bacterial vaccines/adjuvants prevent wheezing episodes in children?

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Purpose of review

To discuss recently discovered mechanisms of action of some bacterial vaccines that may account for their clinical benefit in the prevention of recurrent wheezing and asthma exacerbations in infants and early childhood.

Recent findings

Trained immunity has been shown to confer innate immune cells with a quite long-term nonspecific protection against a broad spectrum of pathogens. Inducers of trained immunity include some bacterial vaccines. Trained immunity-based vaccines (TlbV) of bacterial origin have the capability to induce nonspecific responses to a variety of pathogens, including respiratory viruses, in addition to their nominal bacterial antigens. Clinical data, from epidemiological surveys to well designed randomized clinical trials, indicate that TlbV formulated with bacteria prevent respiratory tract infections of viral cause, such as those associated with recurrent wheezing or asthma exacerbation, in children. Administration of these vaccines by the mucosal route may be important for their outcome in respiratory infections.

Summary

Mucosal bacterial immunotherapy, including certain TlbV, confer protection against a broad spectrum of pathogens, such as viruses, through a mechanism mediated by trained immunity. Clinical studies on the use of these preparations against recurrent wheezing reflect these mechanistic effects. These findings open a new avenue for the development of new strategies for this condition.

Keywords

mucosal immunotherapy, recurrent wheezing, trained immunity, trained immunity-based vaccines, viral respiratory infections

INTRODUCTION

It has become increasingly evident that early-life microbial exposure is key for protection or susceptibility to respiratory diseases later in life. Such an exposure at early stages has an impact on the immune development in the airways, which in turn will modulate the risk of susceptibility to respiratory diseases beyond childhood [1]. Numerous studies in young children document the relationship between lower respiratory tract infections and episodes of wheezing with the development of later asthma [2[•],3]. Therefore, respiratory infections in the first years of life, either the infections themselves and/or their susceptibility, are phenomena related to the susceptibility to develop asthma. In this regard, it is widely accepted that viral infections are particularly important [4], although those of bacterial cause may also play a role [5]. The viruses most commonly involved, both in triggering wheezing episodes

and the onset of asthma itself, are respiratory syncytial virus (RSV) and rhinoviruses [6–8]. With this perspective, it seems reasonable to assume that avoiding early viral infections could prevent wheezing attacks (WA) and asthma exacerbations in children. In this review, we will address recent insights of emerging strategies able to cope with respiratory viral infections using the ability of bacteria to confer resistance against multiple pathogens, especially by inducing trained immunity.

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KEY POINTS

- Viruses play a key role in the inception and acute attacks of asthma (wheeze) in children/infants.
- Innate immunity plays a crucial role in the body's defense against bacterial and viral infections.
- Bacterial PAMPs lead to a potent stimulus on innate immunity that can also be 'trained' to respond faster, broader, and more steadily against reinfections, by inducing stable epigenetic changes on myeloid cells.
- Trained Immunity-based Vaccines (TlbV) can prevent asthma/wheeze viral-induced attacks in children.

STRATEGIES TO PREVENT RESPIRATORY VIRAL INFECTIONS

Effective prevention of respiratory infections because of airborne viruses is not an easy task. We have learned from the current SARS-CoV-2 pandemic that the adoption of widespread personal measures by the entire population has proven to be effective not only against the spread of this virus but also of those that trigger wheezing attacks in children [9]. In the absence of specific vaccines, however, it is not realistic to maintain those preventive measures forever. A Lancet Commission highlights the difficulties of preventing respiratory

viral infections by RSV and rhinoviruses in children [10]. Effective vaccines against RSV are still not available after years of intensive research, and vaccines for rhinoviruses must contend with the challenges faced by the numerous subtypes in circulation [11,12]. In this context, a plausible approach would be to harness the potential of the innate immune system as a defense mechanism against viral infections, such as those based on trained immunity [13] (Fig. 1).

TRAINED IMMUNITY

In later years, it has become clearly evident that innate immune cells may undergo long-term adaptations in response to certain stimuli, which are dependent on persistent epigenetic modifications and metabolic reprogramming of these cells [14,15[¶],16]. This functional adaptation has been termed 'trained immunity', as it confers to innate immune cells an improved performance against multiple pathogens that lasts over time [17,18]. It is, therefore, a type of memory of innate immune cells that, in contrast to adaptive immunity memory, is not antigen-specific [19]. Trained immunity may last up to a year by virtue of epigenetic rewiring induced in long-lived tissue macrophages and in hematopoietic precursors of myeloid cells [20,21]. Interestingly, it can be transmitted across generations in mice models [22[¶]]. Inducers of trained

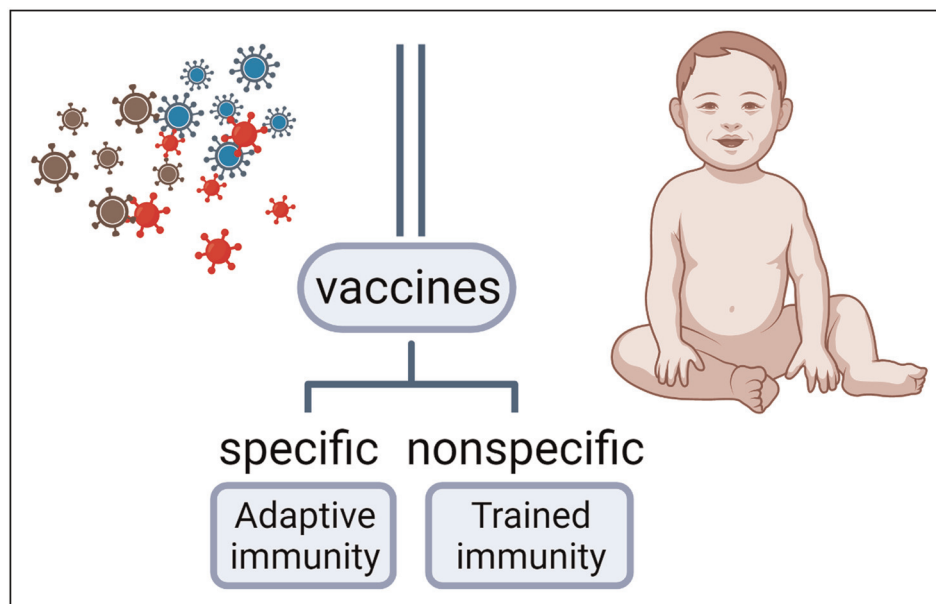


FIGURE 1. Role of bacterial vaccines in the prevention of wheezing. Bacterial vaccines act through different mechanisms inducing a nonspecific immune response (trained immunity) and a specific immune response that leads to protection against a broad spectrum of pathogens. Thus, during infancy and early childhood (first 2–3 years of life), early microbial exposure in the context of vaccination will result in a protection against infection and a decrease in wheezing episodes (this figure was created with BioRender.com.).

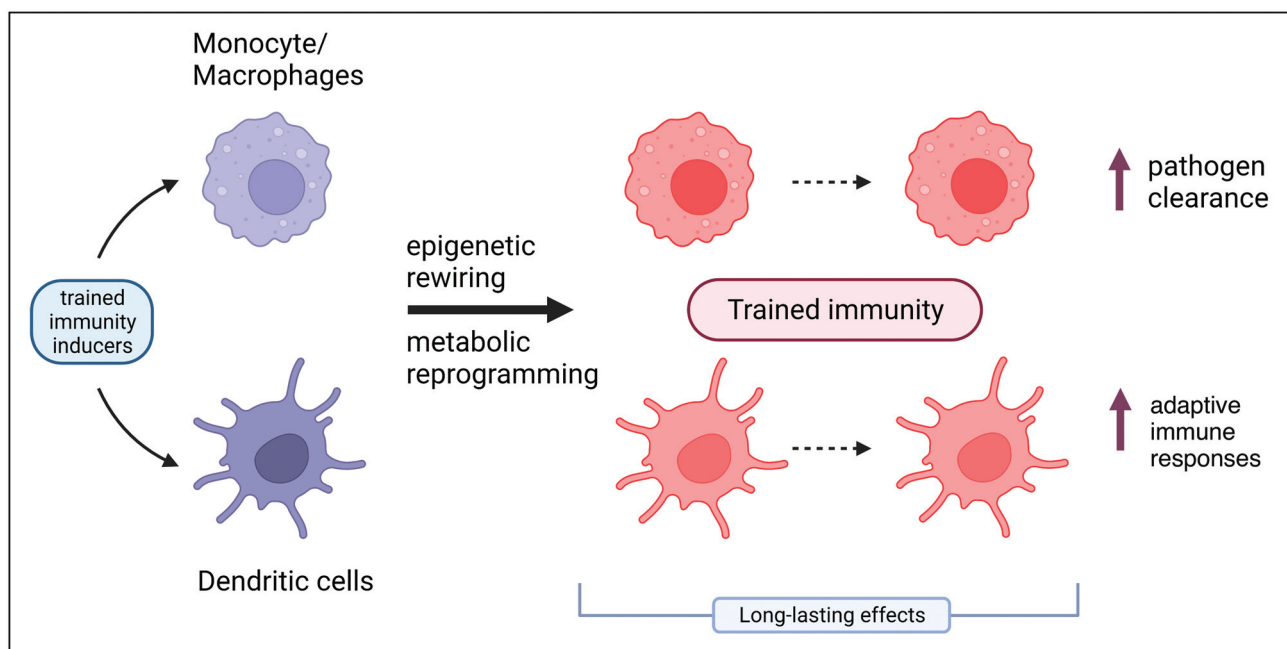


FIGURE 2. Mechanism of action of trained immunity. Trained immunity inducers act on cells from the innate immune system such as monocytes, macrophages and dendritic cells triggering epigenetic rewiring and metabolic reprogramming. These changes will induce trained immunity that, in turn, will contribute to increased pathogen clearance and will enhance adaptive immune responses (this figure was created with BioRender.com).

immunity include microbial components derived from bacteria, fungi or viruses that are ligands of certain PRRs (Pattern Recognition Receptors) present on innate immune cells. Once ‘trained’, the innate cells return to a new resting state ready to act more quickly and with better performance in the event of a subsequent encounter with pathogens. As trained immunity is nonspecific, it confers protection against viruses even if induced by bacteria or fungi and vice versa but also may impact on adaptive responses enhancing T-cell responses [23[¶]] (Fig. 2). Although trained immunity is a physiological mechanism of adaptation of the innate immune system in response to natural infections, it can be induced by certain vaccines and exploited in the design of novel anti-infectious broad-spectrum preparations [24,25,26[¶]].

TRAINED IMMUNITY-BASED VACCINES

By harnessing the properties of trained immunity, a new concept of broad-spectrum vaccines (TibV) has been described [27]. These are defined by two elements, the trained immunity inducers, and the antigens themselves, the former acting as a key stimulant/adjuvant that trains myeloid cells. In this regard, a number of current conventional vaccines have been shown to induce trained immunity, which is considered an explanation for the heterologous protection attributed to these vaccines [28–

30]. The best studied example is the bacterial live-attenuated BCG vaccine from *Mycobacterium bovis* [17,31,32]. By inducing trained immunity, BCG may confer protection not only against virus infections in animal models but also in humans (reviewed by Moorlag *et al.* [33]). In fact, BCG was proposed for the prevention of coronavirus disease 2019 (COVID-19), before specific vaccines became available [34]. This is also the case for MV130, a fully inactivated bacterial mucosal vaccine that, by inducing trained immunity, confers protection against RNA/DNA viruses in respiratory infection models [35^{¶¶}], including SARS-CoV-2 [36^{¶¶}]. Additionally, MV130 has been shown to be effective in preventing respiratory tract infections in different clinical settings [37–41]. Importantly, as the mechanism of protection depends on trained immunity, the effect of these vaccines is long-lasting, that is, as long as trained immunity is present. Thus, bacterial TibV have been shown to confer fairly durable protection against viruses and other pathogens beyond the nominal bacterial antigens they contain [27].

BACTERIAL INDUCERS OF TRAINED IMMUNITY IN THE PREVENTION OF WHEEZING ATTACKS IN CHILDREN

The impact of neonatal BCG vaccination to reduce the prevalence of wheeze and asthma symptoms in children has been documented before the concept

of trained immunity was established. In 2002, a large epidemiological study in German preschool children found a moderate protective effect of BCG vaccination against asthma using standardized questionnaire data obtained at screening for school maturity [42]. This was confirmed by a Spanish study using the International Study of Asthma and Allergies in Childhood questionnaire with a statistically significant lower prevalence of asthma (12-month prevalence of wheeze) for BCG-vaccinated children [43]. A statistically significant association was also found in a study conducted in England between neonatal BCG vaccination and a lower prevalence of wheezing in children aged 6–11 years, showing a 27% reduction [44]. Of note, when this cohort was followed up to adolescence, the protective effect of BCG vaccination was lost, concluding that the results on asthma were relatively short-lived over the first years of life without effects on other atopic diseases [45]. A recent systematic review and meta-analysis of 20 studies and over 220 000 participants may also point in the same direction as the conclusion was that BCG vaccination in early childhood significantly reduced the risk of developing asthma in children not attributable to a reduction in atopic disease [46²²]. Thus, it appears that neonatal BCG vaccination may be reducing wheezing/asthma attacks by conferring protection against lower respiratory infections through a transient induction of trained immunity rather than by reducing allergies.

MV130 also induces trained immunity [35²²], but unlike the intradermal self-replicating BCG vaccine, it is administered daily by the mucosal route and is composed of six fully inactivated whole-cell bacterial species commonly responsible for respiratory tract infections [47]. A randomized clinical trial (RCT) analyzing the efficacy of sublingual MV130 in preventing wheezing attacks in 120 children (under 3 years of age) who had at least three wheezing attacks during the previous year has recently been published [48²²]. Of note, children with positive skin tests for allergens common in their environment were excluded to limit the focus to wheezing episodes directly associated with respiratory infections. Treatment lasted 6 months and patients were evaluated while on treatment and 6 months later, that is, 1 year after the start of treatment. The cumulative number of wheezing attacks during the 1-year period was very significantly reduced in children with active treatment with respect to placebo (main outcome). The same was true for the 6-month period without treatment and for the other secondary variables analyzed (days without wheezing attacks, duration of wheezing attacks, symptom and medication score during attacks or use of healthcare

resources). Thus, the study concludes that MV130 prevents wheezing episodes in young children, most likely caused by respiratory virus infections, even after treatment discontinuation, supporting a durable effect that fits with a mechanism based on trained immunity [48²²].

MUCOSAL IMMUNOTHERAPY WITH BACTERIAL LYSATES IN THE PREVENTION OF WHEEZING/ASTHMA ATTACKS IN CHILDREN

Several RCTs provide evidence that bacterial lysates administered through the mucosa are a useful strategy for the prevention of wheezing/asthma attacks in the pediatric population [49]. Although these products, as with MV130, are mostly derived from bacterial species associated with respiratory tract infections, their ability to induce trained immunity is unknown, with an underlying mechanism(s) of action not fully understood [50,51,52²²].

OM-85 is an oral bacterial extract derived from lysates obtained chemically, which has long been used to prevent recurrent respiratory tract infections in both adults and children. Referring specifically to children, an RCT published in 2010 analyzed the effect of this product in 75 children aged 1–6 years with three or more acute attacks of wheezing in the preceding 6 months [53]. The treatment was for 10 days each month for three consecutive months. The primary endpoint was the total number of wheezing attacks after treatment. At 3 months of treatment, and consistently at 6, 9 and 12 months, the cumulative number of wheezing attacks was significantly lower in the active group as compared with placebo, indicating a durable effect [53]. A more recent RCT, using the same product, attempted to analyze the primary preventive effect of severe respiratory disease in 59 at-risk young children (4–7 months of age) over a two-winter period, treating the first year [54]. No significant differences between active and placebo groups were found in this respect. However, the product appeared to be effective to prevent lower respiratory infections during the first winter, although no carry-over effect could be observed in the second winter [54].

The EOLIA study, an RCT conducted in 152 children with persistent house dust mite-allergic asthma (6–16 years of age), investigated a polyvalent mechanical bacterial lysate administered sublingually for 10 consecutive days in three consecutive months [55]. The primary endpoint of efficacy was improving the asthma control level. Although there were no significant differences at 3, 6 and 9 months in this outcome, the number of

asthma exacerbations per patient was significantly reduced in the active compared with the placebo group after 9 months of follow-up [55].

Two systematic reviews published to date examining this topic suggest that immunomodulatory bacterial compounds are able to reduce the incidence of acute lower respiratory tract disease by up to 40% on average in susceptible children [49,56].

Taken together, the accumulating evidence suggests that mucosal bacterial immunotherapy is beneficial for the prevention of wheezing episodes/asthma exacerbations in children. Given the viral cause of many of these episodes, they are likely be acting by conferring protection against their airway infections. Whether these bacterial immunostimulants are acting as inducers of trained immunity, and/or by alternative mechanisms [51], is not yet clear.

THE POTENTIAL OF BACTERIAL TRAINED IMMUNITY-BASED VACCINES IN RECURRENT WHEEZING

TibV may be acting by conferring nonspecific protection against viral respiratory infections that trigger wheezing attacks or asthma exacerbations in children, as exemplified by MV130 or BCG. As trained immunity is relatively long-lasting, its protection lasts long enough to be effective in a recurrent infectious condition. In addition to offering broad protection against a variety of pathogens, including viruses, trained immunity shows an adjuvant effect that may enhance adaptive responses to the bacterial antigens containing in a given TibV (Fig. 3). Such an enhancement may also be relevant when considering adaptive responses to natural infections. This has been demonstrated in experimental models of immunization with bystander

antigens with MV130 [36²²,47] or with others inducers of trained immunity [57], raising the concept of trained immunity-based adjuvants [27]. In fact, the well known complete Freund's adjuvant does contain heat-killed *Mycobacterium tuberculosis* [58]. Interestingly, this concept is being used in the design of novel RSV–BCG recombinant vaccines [59]. In the case of MV130, the bacterial composition includes *Haemophilus*, *Streptococcus* and *Moraxella*, bacterial species positively associated with lower tract infections [60]. Therefore, a specific response to bacterial antigens is also expected in a bacterial TibV, which may contribute to the overall resistance against pathogens causing respiratory infections. In this regard, T-cell responses against MV130-containing bacteria are readily detected in mice immunized sublingually with MV130 [47].

Another important issue regarding trained immunity-related protection conferred by TibV may be the route of vaccine administration. As the portal of entry of respiratory viruses is the mucosa, training of innate immune cells present in these tissues should be relevant. In this regard, intravenous BCG showed no protection against influenza virus in an experimental model [61]. Although this was not the case by the same route against SARS-CoV-2 in another experimental model, BCG failed to show protection when instead of intravenously it was administered subcutaneously [62]. In this respect, it has been shown that trained immunity is stronger when BCG is administered through the mucosal rather than by the standard intradermal route [63²²]. Thus, it appears that modulation of innate cells through mucosal surfaces is a better approach for the induction of trained immunity. This supports this route for TibV, such as the sublingual MV130, and eventually for the above bacterial lysates used for mucosal immunotherapy.

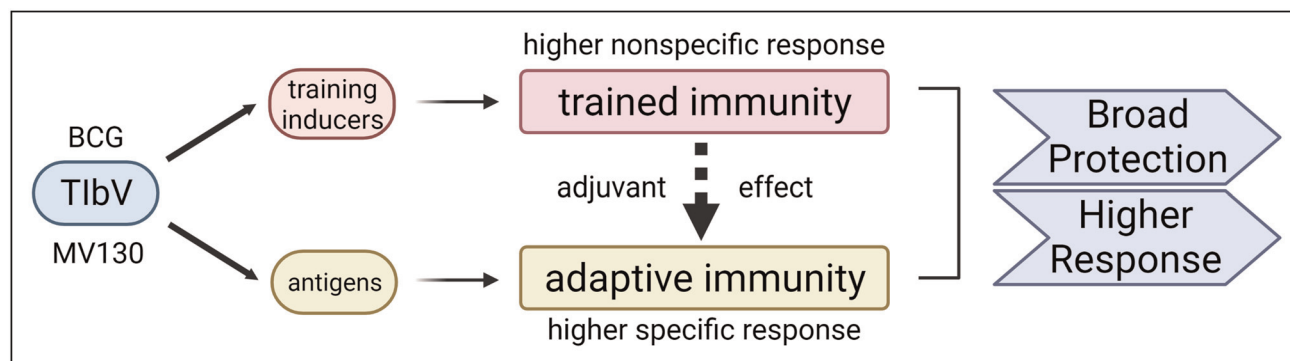


FIGURE 3. Components of trained immunity-based vaccines and mechanism of action. TibV consists of two main components: trained immunity inducers and antigens. The former triggers a highly nonspecific immune response through trained immunity and the latter induces a highly specific response mediated by adaptive immunity. All this leads to broad protection against a wide range of pathogens (this figure was created with BioRender.com).

CONCLUSION

Mucosal immunotherapy with bacterial lysates has been widely used over the last few decades for recurrent respiratory tract infections, with a poorly understood mechanism of action. New concepts emerged in recent years, like trained immunity, showing the potential of innate immune system to prevent respiratory virus infections may give some clues. Recent studies demonstrating that an inactivated whole-cell bacterial sublingual preparation that prevent recurrent wheezing in children induces trained immunity and protects against respiratory viral infections, support such a mechanism shedding light in this regard.

Increased knowledge on trained immunity may be exploited for the design of not only new vaccines such as trained immunity-based vaccines or adjuvants but also to novel vaccination strategies for the primary prevention of wheezing in children at risk. Recent studies providing evidence that trained immunity can be transmitted across generations may point to this direction [22[■]]. In addition, transmission of protection from mothers to offspring against lethal neonatal respiratory rhinovirus infection by treating the former with bacterial lysates may support that view [64[■]].

In summary, the identification and development of new tools with a broad anti-infectious spectrum of protection for improved immunization purposes as T1bV, is a novel and promising strategy for the prevention of respiratory viral infections, including recurrent wheezing in children.

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Conflicts of interest

LC and JLS are employees of Immunotek, the manufacturer of MV130 mentioned in this review. AN and MN declare no conflict of interest.

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